

Emerging Applications of Embedded Systems

The growing ubiquity of low-cost sensors that measure everything from the ambient temperature to people's movements will provide interesting opportunities for embedded systems developers to create innovative smart systems



RICHARD BROWN
VICE PRESIDENT, INTERNATIONAL
MARKETING, VIA TECHNOLOGIES, INC.

If you walk around any new shopping mall in India, you are likely to pass by at least a couple of large screens flashing the latest domestic and international news, cricket results, product advertisements and announcements of special in-store deals and promotions.

This is just the first step in a trend in which screens appear everywhere from your favourite restaurant or store to the railway and bus station; not to mention the backs of the seats in the train carriages and buses putting all relevant information literally at your fingertips. Small, low-power, yet high-performance systems that never break down will be required to process all the information hosted on the cloud and deliver it in multimedia touch-friendly

formats to the screens in increasingly high resolutions. Highly sophisticated software applications will also be required to manage this mountain of information and make sure that you receive exactly the information you need, at the time you need it.

Intelligent digital signage and display systems are just one example of the tremendous growth potential that exists in the embedded market—not just in India but also worldwide. These also provide an exciting opportunity for the Indian embedded industry to leverage its world-class software development expertise with low-power x86 and ARM-based hardware platforms to develop innovative value-added solutions. These solutions can be applied across a wide range of applications and industries including retail, hospitality, travel, entertainment and education—the possibilities are endless!

Putting up screens will be a major trend in homes as well, where there will be an explosion of smart embedded devices for applications such as entertainment, energy management, security and lighting control. The growing ubiquity of low-cost sensors that measure everything from the ambient temperature to people's movements will provide interesting opportunities for embedded systems developers to create innovative smart systems and services that until recently seemed confined to the realms of science fiction.

Imagine, for example, a smart bed that uses sensors to collect information about the time you spent in deep sleep, your body temperature and weight, and your movements

throughout the night. When you wake up, it gives you advice on what to eat for breakfast and what vitamins and nutritional supplements you need to take to start your day on the right foot. Such a device is really not that far away from becoming a reality. In fact, there is a good chance that somebody somewhere in the world is already working on the concept.

With the growing popularity of intelligent smart devices that help you track everything from your heart beat rates, blood pressure and other vital signs to your daily diet and exercise, healthcare and wellness is another huge potential segment for the embedded industry. The opportunities for innovation in this area are amazing.

Smart mobile devices is another segment that holds big promise for embedded systems developers in India. The growing popularity of iPad and low-cost Android tablets is just the first sign of its potential. The proliferation of 3G and ultimately 4G wireless networks will only serve to further boost this momentum. Sales force automation and transportation and logistics management are just two examples of the type of applications that smart mobile devices will be adopted for.

The embedded industry is at the start of a tremendous growth curve as a result of the convergence of a number of key trends including the declining cost of hardware such as screens, growing ubiquity of wireless broadband, emergence of the cloud, and the development of lower-power yet higher-performance ARM system-on-chips and x86 processor platforms. ●